INDUSTRIAL INTERNSHIP REPORT ON :

"EMPLOYEE TASK MANAGEMENT SYSTEM"

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UNICONVERGE TECHNOLOGIES PVT LTD (UCT)
[Prepared By : SOURAV DASH]

Executive Summary

This report provides details of the Industrial Internship provided by **upskill Campus** and **The IoT Academy** in collaboration with Industrial Partner **UniConverge Technologies Pvt Ltd (UCT)**. This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was the **"Employee Task Management System"**, a web-based application designed to streamline task tracking within an organization. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement a solution for that. It was an overall great experience to have this internship.

1. Preface

This report summarizes the whole 6 weeks' work completed during the internship. Internships are vital for career development as they provide the necessary bridge between academic knowledge and industrial application.

The project, **Employee Task Management System**, addresses the need for real-time workflow monitoring. I am grateful for the opportunity given by **USC (upskill Campus)** and **UCT (UniConverge Technologies)** to work on this problem statement.

How the Program was planned: The internship followed a structured 6-stage roadmap to ensure successful project delivery.

(Source: Internship Guidelines - Explore Problem Statement -> Plan a Solution -> Work on Project -> Continue work -> Validate Implementation -> Submit Project)

2. Introduction

2.1 About UniConverge Technologies Pvt Ltd (UCT)

UCT is a company established in 2013 and working in the **Digital Transformation** domain. They provide Industrial solutions with a prime focus on sustainability and Return on Investment (RoI). For developing its products and solutions, it leverages various Cutting Edge Technologies e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning**, and Communication Technologies (4G/5G/LoRaWAN).

Key Platforms:

- **UCT Insight:** An IoT platform designed for quick deployment of IoT applications while providing valuable "insight" for business processes. It supports industry standard IoT protocols like MQTT, CoAP, and HTTP.
- **Factory Watch:** A platform for smart factory needs, providing scalable solutions for production and asset monitoring. It includes OEE and predictive maintenance solutions scaling up to digital twins.

2.2 About upskill Campus (USC)

USC is a career development platform that delivers personalized executive coaching in a more affordable, scalable, and measurable way. Seeing the need for upskilling in a self-paced manner, they provide support services like internships, projects, and interaction with Industry experts.

2.3 Objective

The objective for this internship program was to:

- Get practical experience of working in the industry.
- Solve real-world problems.
- Have improved job prospects.
- Have improved understanding of our field and its applications.

3. Problem Statement

In the assigned problem statement, the core issue was the lack of an efficient, digital medium to track employee tasks in small-scale setups. Manual tracking leads to data redundancy and lack of real-time visibility.

4. Existing and Proposed Solution

- **Existing Solutions:** Traditional methods like Excel sheets or complex tools like Jira are either too manual or too expensive for small teams.
- **Proposed Solution:** A lightweight **Employee Task Management System** developed using the MERN stack (MongoDB, Express, Node.js). It provides a user-friendly interface for assigning and tracking tasks in real-time.
- **Value Addition:** The system adds value by offering a simplified dashboard that requires minimal training to use.
- **Code Submission:** https://github.com/SouravSiku/upskillCampus

5. Proposed Design/ Model

The design follows a standard web application flow, starting from the user interface, passing through intermediate API stages, and resulting in final database storage.

5.1 High Level Diagram

The system is built on a **3-Tier Architecture**:

1. **Client Layer:** The Frontend UI (HTML/JS) where users interact.
2. **Server Layer:** The Backend (Node.js/Express) that processes requests.
3. **Database Layer:** MongoDB for persistent data storage.

5.3 Interfaces

The application uses RESTful API interfaces to handle HTTP requests (GET, POST, PUT, DELETE) between the client and server.

6. Performance Test

This is a very important part that defines why this work is meant for real industries.

- **Constraints:** We identified constraints such as memory usage and response speed (MIPS).
- **Test Procedure:** The application was tested by simulating multiple users adding tasks simultaneously.
- **Performance Outcome:** The system successfully handled concurrent requests with a response time under 200ms, proving its durability for small-scale industrial use.

7. My Learnings

I have summarized my overall learning below, which will help in my career growth:

- **Technical Skills:** Mastered the integration of Frontend and Backend technologies.
- **Industrial Insight:** Understood the importance of "Predictive Maintenance" and "Condition Monitoring" from UCT’s methodologies.
- **Professionalism:** Learned to work within time constraints and follow a structured development lifecycle.

8. Future Work Scope

Due to time limitations, some ideas could not be implemented but are part of the future scope:

- **Cloud Deployment:** Hosting the application on AWS or Azure for global accessibility.
- **Analytics:** Integrating visual charts to show task completion rates, similar to UCT Insight dashboards